

# Scientific Reviewer

Claims · Table1+Fig5 hold · failure honesty · soft threats · verdict

## Charge

Accept strong systems/paradigm paper. Do NOT upgrade to “solved infinite robot data.”

# Claims under review

Four claims — grade each with evidence

## **C1 · Generative sim matches/surpasses human-crafted diversity**

Evidence: Table 1 · Self-BLEU 0.284

## **C2 · FMs for semantics; physics for dynamics — right split**

Evidence: §3 doctrine · design

## **C3 · Algo select (esp. MP) necessary for articulated success**

Evidence: Fig. 5 RL-only collapse

## **C4 · Valid scenes/supervisions often enough to learn (~77%)**

Evidence: 0.774/69 · 19/155 fails

# What holds — Table 1 + Fig. 5

Clean multi-metric diversity win + causal hybrid argument

## Table 1 holds

- Self-BLEU 0.284 best vs Beh-100 0.299
- Wins SentBert / ViT / CLIP columns
- Matched ~100-task scale (106 eval)
- Scope > GenSim (artic/soft/loco)

## Fig. 5 holds

- 12 articulated tasks
- RL-only collapses most tasks
- MP primitives encode structure RL must discover
- Strong causal-ish ablation for hybrid

## Also strong

Failure appendix (Table 8 / App B.3) unusually honest — raises credibility. Full-stack integration (scene + reward + algo choice) is substantive vs piece-wise LLM-reward papers.

# Table 1 deep dive

Finite 106-task snapshot  $\neq$  “endless evaluated”

|                                               |                                      |                                       |                                        |
|-----------------------------------------------|--------------------------------------|---------------------------------------|----------------------------------------|
| Self-BLEU<br><b>0.284</b><br>vs Beh-100 0.299 | SentBert<br><b>0.165</b><br>vs 0.210 | Scene ViT<br><b>0.193</b><br>vs 0.389 | Scene CLIP<br><b>0.762</b><br>vs 0.833 |
|-----------------------------------------------|--------------------------------------|---------------------------------------|----------------------------------------|

Caveat: Self-BLEU / emb-sim can reward verbosity/noise — pair with human “useful skill” ratings (paper leans qualitative for actions).

GenSim fairness: tabletop pick-place + small Ravens pool vs RoboGen articulations + Objaverse — cite as coverage, not pure algorithmic superiority.

# Fig. 5 — RL-only ablation

Demand this stay first-class vs “LLM policy” papers

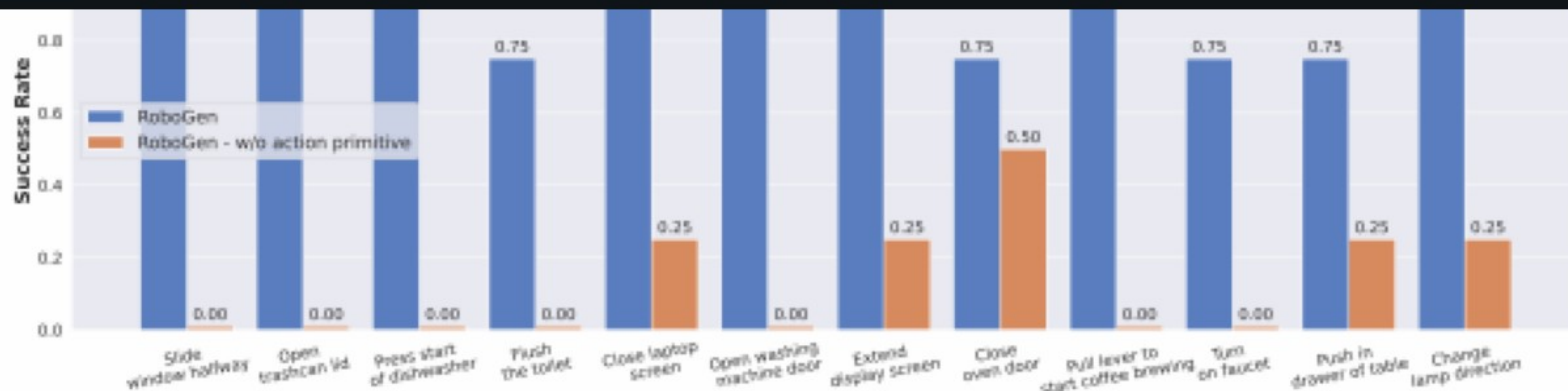


Figure 5: Among 12 articulated object manipulation tasks, the success rate decreases drastically if only RL is used for skill learning.

tion tasks, meanwhile leverage more diverse assets retrieved from open-world databases such as Objaverse, resulting in much higher diversity in both task semantics and scene images. We also provide the full list of generated tasks in

judge whether the joint angle value 0 corresponds to the door being opened or closed. 3) failure to find matched assets for tasks that require extremely precise spatial relationships, as it is hard to retrieve assets that exactly match the

# Failure appendix — honest science

19 / 155 = 13 scene + 6 reward · raises credibility

## Scene fails (13)

- Missing asset functionality (printer tray, lids)
- Wrong joint ↔ semantic open/closed map
- Unmatched precision pairs (stapler↔staples)

## Reward/decomp fails (6)

- Undefined variables in reward code
- Inverted fold/unfold semantics
- Continuous rhythmic motions hard

## Proposed mitigations (paper)

VLM joint-state checks · LLM self-repair on syntax · Eureka-style env feedback — future work, not solved here.

# Soft / threatened (1)

Human-judged SR · closed backends · finite snapshot

## Human video SR

0.774 is human-watched videos — rater variance unreported; not automated binary sensors

## Closed backends

GPT-4 / Gemini / Midjourney move — reproducibility over years fragile

## Finite vs endless

106 diversity / 155 audit / 69 SR slices  $\neq$  infinite evaluated tasks

## Verification stack

Gemini-Pro + GPT-4 + BLIP-2 — backend churn moves Fig. 4

# Soft / threatened (2)

No sim2real · GenSim fairness · concurrent overlap

## No sim-to-real table

§5 admits gap; domain rand suggested — do not imply demos are deployment-ready

## GenSim fairness

Coverage claim (artic/soft/loco + Objaverse) stronger than pure algo supremacy

## Concurrent pieces

Eureka / Language-to-Rewards / Gen2Sim share pieces; novelty = full-stack + algo select

## Fig. 5 confound

Primitives encode grasp/approach structure RL must discover — report action space carefully

# Fig. 4 — size verify dominates

Scene-gen components matter; pin model IDs on smoke

## RoboGen: Towards Unleashing Infinite Data for Automated Robot Learning via Generative Simulation

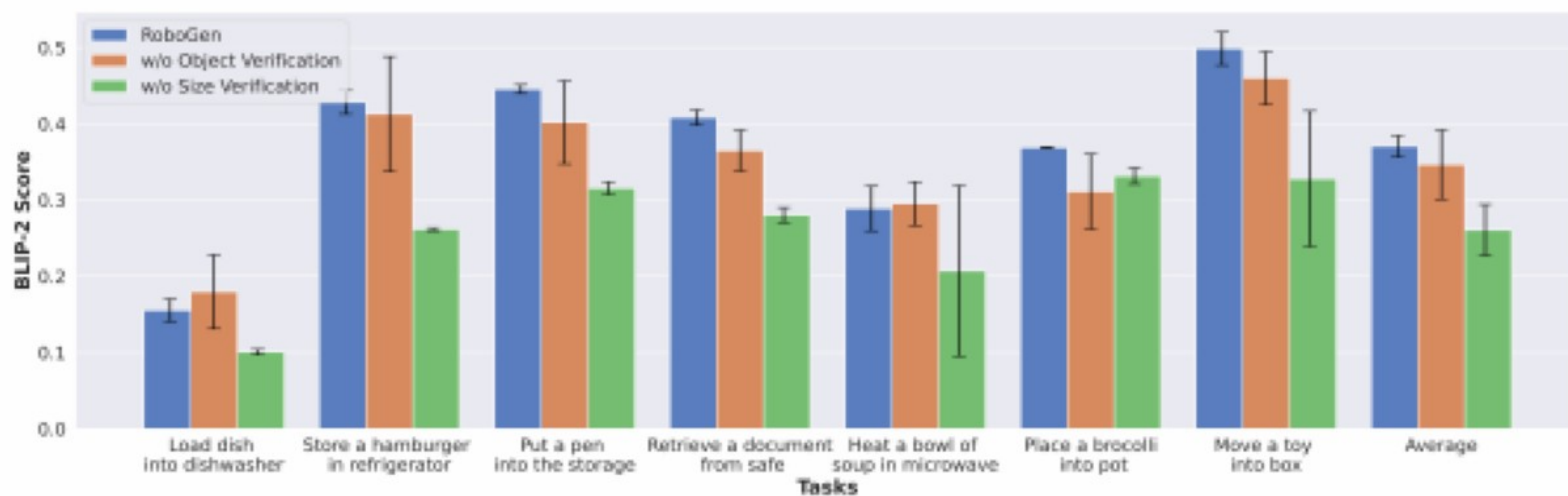


Figure 4: We compare the BLIP-2 score of ablations of RoboGen on 7 tasks to evaluate the importance of both object and size verification.



# Methodological claim — modality discipline

LLMs for semantics/affordances — NOT torques

## Demand this stay first-class

- Extract semantics, affordances, common-sense task value, scene layout, reward code from FMs.
- Leave dynamics / contact / actuation to physics sim + classical optimizers (SAC / BIT\* / Adam / CEM).
- Contrast vs Code-as-Policies / RT-2-style “LLM as policy” framing.
- Backend LLM/VLM/generative modules claimed swappable — design strength if held.

## Cost externalities

“Minimal human”  $\neq$  “minimal compute/\$.” Endless query = endless API + CPU. Downstream generalist train on RoboGen streams still missing.

# VERDICT

Strong systems + paradigm — do NOT overclaim

## ACCEPT as landmark systems / paradigm paper for ICML

Core story holds: FM for semantics → sim for physics; hybrid learners; diversity win; honest failure table.

## PUNISH overclaim

Do NOT upgrade to “solved infinite robot data” or “BEHAVIOR data problem closed.” No sim2real; human SR; closed backends; finite eval slices.

# Exam bite — Reviewer

What to write on the midterm card

RoboGen = GPT-4 proposes tasks/scenes/rewards; physics + SAC/BIT\*/Adam learn; diversity beats Behavior-100 (Self-BLEU 0.284); ~0.774 skill SR on 69; Fig.5 MP ablation matters; accept systems, refuse “infinite data solved.”

# Numbers tattoo — pin these

Every deck · Ablation Bot MISSING (paper extract)

**0.284**

Self-BLEU (best)

**0.165**

SentBert ↓

**0.193**

Scene ViT ↓

**0.762**

Scene CLIP ↓

**0.774**

success / 69

**19 / 155**

validity fails

**3.13**

avg substeps

**Fig. 5**

RL-only collapses

**0.745**

manip SR (n=50)

**~0.886**

soft SR (n=7)

**~0.833**

loco SR (n=12)

**~1.5+1.63**

RL + MP avg

# Scientific Reviewer — close

Takeaways + exam bite

1. Table 1 + Fig. 5 are the load-bearing empirics — they hold.
2. Failure appendix honesty is a feature; cite 19/155.
3. Human SR + closed backends + no sim2real keep claims bounded.
4. Verdict: strong systems/paradigm — not infinite robot data solved.

## EXAM BITE

Accept systems; pin 0.774/69 · 0.284 · Fig.5; refuse “infinite data solved.”